Network Vulnerability Assessment Report

23th SEPT, 2024

Level 2 Report

1.1 Copyright:

The copyright in this work is vested in Cybertryzub Infosec Pvt. Ltd., and the document is issued in confidence for the purpose for which it is supplied. It must not be reproduced in whole or in part or used for tendering or manufacturing purposes except under agreement or with the consent in writing of Cybertryzub Infosec Pvt. Ltd. and then only on condition that this notice is included in any such reproduction. No information as to the contents or subject matter of this document or any part thereof arising directly or indirectly there from shall be given orally or in writing or communicated in any manner whatsoever to any third party being an individual firm or company or any employee thereof without the prior consent in writing of Cybertryzub Infosec Pvt. Ltd.

© Cybertryzub Infosec Pvt. Ltd.

1.2 Contact Information:

Mrs. Jyoti Singh
CISA, CIPM, FIP, CIPP/E, ISO 27001
LA, DevOps, MSc, PG Data Science.
(InfoSec Lead Auditor)
+91 9873784553, 8295777275
info@cybertryzub.com

Corporate Office
PLOT NO 76D UDYOG VIHAR 4 SECTOR 18,
GURGAON – 122001
HARYANA, India

Registered Office
F-300, SECTOR 63, NOIDA-201301
UTTAR PRADESH, INDIA

2.0 About Us:

Cybertryzub InfoSec Private Limited is an ISO Certified (ISO 27001:2013) company. Cybertryzub aims to provide Auditing; Consulting, Information Risk Management and Managed services tailored as per the market requirements and enhance the satisfaction and confidence index of its existing and potential customers.

Cybertryzub has been a CERT-In empanelled IT Security Auditor. It is an acknowledgement of Cybertryzub's technical expertise in conducting Information Security Audits. As a CERT-In empanelled auditor Cybertryzub is qualified to conduct security audits of websites, networks & applications. On successfully completing the audit as per CERT-In Guidelines, our team can issue the the CERT-In Certification as required by compliance requirements.

From the development of a Security Policy, Security Awareness Training, through to the delivery of complete end-to-end solutions that encompass Perimeter Security, Secure Content Management, Identity and Access Management, Vulnerability Assessment, Risk, Policy and Compliance Management, Cybertryzub InfoSec Private Limited helps organizations understand, monitor and mitigate the risks in their IT infrastructure.

Cybertryzub InfoSec Private Limited has information security experts, compliance professionals and process consultants from different industry verticals with multiple certifications each with experience in handling consulting assignments, audits and training programs.

Cybertryzub InfoSec Consultants has established a reputation for providing practical solutions that are both business driven and cost-effective. This has enabled the company to secure the IT Infrastructure of leading Stock Brokers, Government undertakings, Banks, Insurance Companies and Financial Institutions, BPO, KPO, PKI Industry, Data Center, Software Company, Automobile, Healthcare & Life Sciences, Hospitality, Travel, Transportation, Consumer & Retail, Technology, Media & Telecommunication, School, College, Smart City. Security Leadership our experienced team of ethical hackers have identified security vulnerabilities, we have put together a highly qualified team of security researchers with credentials like CISA, CISSP, OSCP, CSOE, CCNP, MCSE+, ISO 27001 LA, ITIL, COBIT, CIPP/E, CIPM, FIS, DISA, CPA, Cyber Law, Data Science, Artificial Intelligence & Machine Learning, DevSecOps, AWS Architect, Azure Architect.

3.0 Table of Contents

1.0 1.1 Copyright 2
1.2 Contact Information.....2
2.0 About Us.....3
3.0 Table of Contents.....4
4.0 Assessment Information.....5
4.1 Assessment scope summary.....5
4.2 Client details.....5
4.3 Cybertryzub Infosec Pvt. Ltd. Details.....5
5.0 Engagement Overview.....6
6.0 Process and Methodology.....7
7.0 Vulnerability Overview.....8
8.0 Scope.....9
9.0 Open Ports.....11
10.0 Summary Of Findings.....12
11.0 Executive Summary Of Findings 13
12.0 Detailed Findings and Recommendations 14
13.0 Tools and References.....15

4.0 Assessment Information

4.1 Assessment Scope Summary

Engagement Type	Gray Box Assessment
Engagement Scope	7 IPs Internal Network Security Assessment
Engagement Timeframe	08/08/2024 to 23/09/2024

4.2 Client Details

Organization	Crego
Contact Person	Mr Abhishek

4.3 Cybertryzub Infosec Pvt. Ltd. Details

Auditor	Deepanshu Kumar
Review and Final Approval	Mrs. Jyoti Singh (CISA, CIPM, FIP, CIPP/E, ISO 27001 LA, DevOps, MSc, PG Data Science.)
Project ID:	
Report Type	Closure Report

5.0 Engagement Overview

Crego has engaged with **Cybertryzub** to conduct a penetration test of their **Public IP/Server**. This report contains all the results of the report as well as all the action items that were included in the penetration test. The purpose of this report is to present the current security level of the external perimeters including gaps, vulnerabilities, and misconfigurations. The findings presented in this report should be fixed to improve the security level of the network systems.

Service Description

Vulnerability Assessment Testing is the process of simulating real-world attacks by using the same techniques as malicious hackers. For a security assessment that goes beyond a simple vulnerability scanner, you need experts in the industry. **Cybertryzub** conducts its penetration test by approaching the scope with both a manual and automatic approach.

Network Penetration Test

Network penetration testing assesses the external perimeter of an organization from the perspective of an attacker that has some or no prior information about the external perimeter. The goal of the penetration test is to assess the external perimeter and expose security gaps, security issues, and misconfigurations that might expose the organization to security risks and attacks.

Project Objectives

Cybertryzub consultants conduct all testing manually combined with custom and commercial tools that perform unique attack approaches on the network to make sure we cover the whole system in the test. Our expert knowledge and experience are the value we provide in our services.

6.0 Process and Methodology

Cybertryzub uses the same techniques as sophisticated real-world adversaries to assess your system and provide unique visibility into security risks that automated approaches often miss. We use a methodology that has been developed out of years of experience and thousands of assessments. This methodology is also based on the globally recognized methodology of OWASP | PTES | OSINT | CWE for network security testing. Our methodology follows these steps:

1 Reconnaissance

This process begins with a detailed scanning and enumeration of the network system and infrastructure, and the information related to the system that is exposed on the internet. After this stage, we conduct manual testing of the gathered data to be analyzed further for attack paths.

2 Automated Approach

Once the scope has been fully enumerated, we start to approach the scope manually and with automatic approaches both self-developed and commercial. The goal is to cover most of the network.

3 Exploitation And Manual Testing

At this stage of the assessment, our experts review the gathered attack vectors and try to exploit the vulnerabilities found in a manual safely approach.

4 Assessment Report

Once the engagement has been completed, we start to develop the assessment report which consists of an executive part and a technical part where we present all the findings and recommendations for fix.

4 Re-test

As an additional option to our standard assessment, we offer a full re-test to our clients to check whether the issues has been fixed by the client and if it’s possible to bypass the patches that were implemented. This will then be updated by a new re-test report

7.0 Vulnerability Overview

Cybertryzub performed a Network Security Assessment for **Crego** on **08/08/2024 –** 23/09**/2024** . This assessment utilized both commercial and proprietary tools for the initial mapping and reconnaissance of the network system as well as custom tools and scripts for unique vulnerabilities. Several attacking tools was used in order to conduct the penetration testing on the network.

VULNERABILITY RISK DEFINITION AND CRITERIA

The risk rating assigned to each vulnerability are determined by averaging several aspects of the exploit and the environment, including reputation, difficulty and impact.

Critical — Critical vulnerabilities pose a serious threat to an organization's security, and should be fixed immediately. They may provide a total compromise of the target environment, or similar critical impacts.

High — High risk vulnerabilities provide a serious risk to the company environment and should be corrected promptly. These issues can significantly affect the organization's security posture.

Medium — Medium severity vulnerabilities represent a moderate risk to the environment. They may require additional context before remediation but should be remediated after critical and high risks.

Low — Low severity vulnerabilities provide minimal risk to the target environment, and often theoretical in nature. Remediation of low risks is often a lower priority than other security hardening techniques.

Informational — Informational vulnerabilities have little-or-no impact to the target scope by themselves. They are included however, as they may be a risk when combined with other circumstances or technologies not currently in place. Remediation of informational items is not necessary.

8.0 Scope

Identified Hosts

The hosts identified during this vulnerability scan is presented in the table below.

In total **7 hosts** were identified during the scan.

IP	FQDN	Operating System	NetBIOS Name	MAC Address
10.30.0.1				
10.30.0.2		Linux Kernel 2.6		
10.30.0.195		Linux Kernel 2.6		
10.30.6.86		Ubuntu 14.04 Linux Kernel 3.13		
10.30.7.207		Linux Kernel 2.6		
10.30.9.171		Ubuntu 14.04 Linux Kernel 3.13		
10.30.11.121		Ubuntu 14.04 Linux Kernel 3.13		

9.0 Open Ports

Host IP	Open TCP Ports	Open UDP Ports
10.30.0.1		
10.30.0.2	53	
10.30.0.195	22,80,443	
10.30.6.86	80	
10.30.7.207	443	
10.30.9.171	80	
10.30.11.121	80	

10. Summary Of Findings

10.1 Overview

The table below displays the number of vulnerabilities identified for each host, during this vulnerability scan.

IP	Critical	High	Medium	Low
10.30.0.1	0	0	0	0
10.30.0.195	0	0	0	1
10.30.0.2	0	0	0	0
10.30.11.121	0	0	0	0
10.30.6.86	0	0	0	0
10.30.7.207	0	0	0	0
10.30.9.171	0	0	0	0

11. Executive Summary Of Findings

Sr. No	Vulnerability Name	Severity	Level 1 25/07/2024	Level 2 23/09/2024
1	ICMP Timestamp Request Remote Date Disclosure	Low	OPEN	Closed

12. Detailed Findings and Recommendations

Vulnerability	ICMP Timestamp Request Remote Date Disclosure				
Synopsis	It is possible to determine the exact time set on the remote host.				
Severity	Low				
Status	**Closed**				

13. Tools and References

TOOLS:

The following tools were used for Security Testing.

- **Nexpose**
- **Kali Linux**
- **Metasploit**
- **Nmap**
- **Nessus**
- **Some other proprietary scripts and tools were also used.**

REFERENCES:

For application security visit www.owasp.org

http://cert.org/other_sources/tool_sources.html

http://securityfocus.com/

http://projects.webappsec.org/w/page/13246978/Threat Classification